

Research Methodology in CS (CSEG3060_3)

Lecture 5: International Scenario of Intellectual Property: Overview of international IP frameworks (TRIPS, Paris Convention, Berne Convention), global trends in patenting, and harmonization of IP laws.

Unit 5: Intellectual Property and Patents • Mapped Course Outcome: CO3

Lecture 5: International Scenario of Intellectual Property

Course: Research Methodology in CS (CSEG3060_3)

Program: B.Tech (Computer Science) | Semester 5

Unit 5: Intellectual Property and Patents

Mapped Course Outcome: CO3

1. Learning Objectives

By the end of this lecture, students should be able to:

- Describe the historical evolution of international intellectual property frameworks and explain why cross-border IP cooperation became necessary in a globalized economy.
- Identify and differentiate the principal international IP treaties, namely the Paris Convention, the Berne Convention, and the TRIPS Agreement, with respect to their scope, signatories, and substantive provisions.
- Analyze the role of the World Intellectual Property Organization (WIPO) and the World Trade Organization (WTO) in administering and enforcing international IP standards.
- Evaluate contemporary global trends in patent filing activity, including geographic concentration, sectoral distribution, and emerging challenges for developing economies.
- Critically assess the ongoing efforts toward harmonization of IP laws and the tensions between uniformity and national sovereignty.
- Apply the principles discussed to real-world case scenarios involving cross-border IP disputes and patent strategies.

2. Introduction

Intellectual Property (IP) has historically been regarded as a territorial right, meaning that protection granted in one country does not automatically extend to another. However, the realities of the modern economy, where goods, services, information, and innovation routinely cross national borders, demanded a coordinated international response. Inventors, authors, and creators could not realistically file separate applications in every country where they sought protection without enormous cost and procedural complexity.

The international scenario of IP therefore concerns the treaties, organizations, norms, and institutions that govern how intellectual property rights are recognized, protected, and enforced across jurisdictions. This lecture examines three foundational instruments that shape this global architecture:

1. The **Paris Convention for the Protection of Industrial Property** (1883)
2. The **Berne Convention for the Protection of Literary and Artistic Works** (1886)
3. The **Agreement on Trade-Related Aspects of Intellectual Property Rights (TRIPS)** (1994)

We will also explore contemporary patenting trends, the institutions that administer these frameworks, and the complex push toward harmonization of IP law across diverse legal traditions.

3. Historical Context: The Need for International IP Cooperation

3.1 The Pre-Convention Era

Before the late nineteenth century, IP was almost exclusively a matter of national law. The Venetian Patent Statute of 1474 and the English Statute of Anne (1710) were early national milestones, but no international recognition existed. An inventor securing a patent in France, for instance, received no protection in Britain, Germany, or the United States.

This created several problems:

- **Incentive distortion:** Innovators with limited resources could not afford protection in all relevant markets.
- **Piracy and counterfeiting:** Foreign markets often became free-rider zones where local competitors could legally copy inventions and creative works.
- **Trade frictions:** Disputes over copied goods created diplomatic tensions, particularly during the Industrial Revolution when European manufactured goods dominated global trade.

3.2 The Industrial Revolution as Catalyst

The Second Industrial Revolution (roughly 1870–1914) intensified these pressures. Mass production, expanding railway networks, and international exhibitions (such as the 1873 Vienna World Exhibition, where Austrian exhibitors complained of piracy of their inventions) created urgent political will for international cooperation. This led directly to the Paris Convention of 1883.

3.3 Twentieth-Century Expansion

The twentieth century saw the proliferation of international IP instruments, the establishment of WIPO in 1967, the inclusion of IP within the GATT framework leading to TRIPS in 1994, and the proliferation of regional agreements such as the European Patent Convention (1973) and the Patent Cooperation Treaty (1970).

4. Major International IP Frameworks

4.1 The Paris Convention for the Protection of Industrial Property (1883)

4.1.1 Background

The Paris Convention, signed on 20 March 1883, was the first major international treaty designed to protect industrial property. It was originally adopted by 11 states, including Belgium, Brazil, France, Guatemala, Italy, the Netherlands, Portugal, El Salvador, Serbia, Spain, and Switzerland. As of recent counts, it has over 180 contracting parties, making it one of the most widely adhered-to international IP treaties.

4.1.2 Scope

The Convention applies to industrial property in its broadest sense, including:

- Patents (inventions)
- Utility models
- Industrial designs
- Trademarks
- Service marks
- Trade names
- Geographical indications (appellations of origin)
- Repression of unfair competition

4.1.3 Core Principles

a) National Treatment (Article 2) Foreign nationals of any member state must be afforded the same protection as nationals of the country where protection is sought. This is a cornerstone principle that prevents discrimination based on nationality.

b) Right of Priority (Article 4) An applicant who files a patent (or other industrial property right) in one member state has a priority period during which subsequent filings in other member states will be treated as if filed on the original date:

Type of Right	Priority Period
Patents and utility models	12 months
Industrial designs	6 months
Trademarks	6 months

This means an inventor can file a single application in their home country and then have 12 months to decide on foreign filings without losing the original filing date. This principle dramatically reduces cost and risk.

c) Independence of Patents (Article 4bis) Patents obtained in different member states are independent of each other. The grant, scope, and termination of a patent in one country does not affect its status in another.

d) Common Rules on Patents The Convention sets minimum standards, including:

- Patents may not be denied solely because the product is imported.
- Compulsory licensing is permitted under certain conditions to prevent abuse.
- The owner of a patent on a product has the right to prevent importation of products manufactured without consent.

4.1.4 Limitations

The Paris Convention sets only broad principles and minimum standards. It does not provide a system for international registration or examination. Substantive patent law (such as what is patentable, examination procedures, term of protection) remains governed by national law.

4.2 The Berne Convention for the Protection of Literary and Artistic Works (1886)

4.2.1 Background

While the Paris Convention addressed industrial property, the protection of literary and artistic works required a parallel framework. The Berne Convention, completed in 1886 and revised several times (most recently in Paris in 1971), provides this framework. It is administered by WIPO and currently has over 180 contracting parties.

4.2.2 Scope

The Convention protects "literary and artistic works," including:

- Books, pamphlets, and other writings
- Lectures, addresses, sermons
- Dramatic and dramatico-musical works
- Choreographic works and entertainments in dumb show
- Musical compositions with or without words
- Cinematographic works
- Works of drawing, painting, architecture, sculpture, engraving, and lithography
- Photographic works
- Applied art works
- Translations, adaptations, and arrangements

Computer programs are explicitly included under "literary works" in many jurisdictions implementing the Berne Convention, a position affirmed by the WIPO Model Provisions and TRIPS.

4.2.3 Core Principles

a) Automatic Protection (Article 5) Protection is automatic and does not require registration, formality, or deposit. Once a work is created and fixed in a tangible medium, it qualifies for protection in all Berne member states without any procedural steps. This contrasts sharply with patent law, where registration is required.

b) National Treatment (Article 5) Works originating in one member state must receive the same protection in other member states as works of national origin.

c) Independence of Protection (Article 5) Protection is independent of whether the work is protected in its country of origin. Even if a work enters the public domain in its country of origin, it may still be protected abroad.

d) Minimum Standards of Protection

The Berne Convention establishes three categories of protected economic rights and moral rights:

Economic Rights:

- Right of translation (Article 8)
- Right of reproduction (Article 9)
- Right of public performance and communication (Article 11)
- Right of broadcasting (Article 11bis)
- Right of recitation (Article 11ter)
- Right of adaptation and arrangement (Article 12)

Moral Rights (Article 6bis):

- Right of paternity (right to claim authorship)
- Right of integrity (right to object to distortion or modification harmful to honor or reputation)

e) Duration of Protection (Article 7) The general rule is the life of the author plus 50 years. Some categories receive longer terms, and many member states now provide life plus 70 years.

4.2.3 The Three-Step Test (Article 9(2))

Reproduction rights are subject to a limitation known as the three-step test, which has become foundational in international copyright law:

1. Limitations and exceptions must be confined to **certain special cases**.
2. They must not **conflict with a normal exploitation of the work**.
3. They must not **unreasonably prejudice the legitimate interests of the author**.

This test has been transposed into TRIPS and the WIPO Copyright Treaty, influencing global copyright doctrine.

4.3 The TRIPS Agreement (1994)

4.3.1 Background

The Agreement on Trade-Related Aspects of Intellectual Property Rights (TRIPS) was negotiated during the Uruguay Round of the General Agreement on Tariffs and Trade (GATT) and entered into force on 1 January 1995 as part of the Marrakesh Agreement that established the World Trade Organization.

TRIPS represented a paradigm shift in international IP law. Unlike the Paris and Berne Conventions, which were administered by WIPO and focused on cooperation among IP offices, TRIPS was a trade agreement with binding enforcement mechanisms through the WTO dispute settlement system.

4.3.2 Objectives (Article 7)

TRIPS pursues objectives such as the reduction of distortions and impediments to international trade, the promotion of effective and adequate protection of IP rights, and the assurance that measures to enforce IP do not themselves become barriers to legitimate trade.

4.3.3 Core Obligations

a) National Treatment and Most-Favoured-Nation Treatment TRIPS obliges all WTO members to grant:

- National treatment to foreign IP right holders
- Most-favoured-nation treatment (any advantage granted to nationals of one member must be extended to all members)

b) Minimum Standards of Protection

TRIPS sets minimum standards across multiple IP domains:

IP Domain	TRIPS Provisions
Copyright and Related Rights	Articles 9–14
Trademarks	Articles 15–21
Geographical Indications	Articles 22–24
Industrial Designs	Articles 25–26
Patents	Articles 27–34
Layout Designs of Integrated Circuits	Articles 35–38
Undisclosed Information (Trade Secrets)	Articles 39–40
Anti-Competitive Practices	Article 40

c) Patent Provisions (Articles 27–34)

Key patent obligations under TRIPS include:

- **Patentable Subject Matter:** Patents must be available for inventions in all fields of technology, whether products or processes, provided they are new, involve an inventive step, and are capable of industrial application (Article 27(1)).
- **Non-Discrimination:** Patents must be granted without discrimination as to place of invention, field of technology, or whether products are imported or locally produced (Article 27(1)).
- **Term of Protection:** Patent term shall not end before 20 years from the filing date (Article 33).
- **Compulsory Licensing:** Permitted under specific conditions (Article 31), including for public non-commercial use, anti-competitive practices, or dependent patents.

d) Enforcement Provisions (Part III)

TRIPS requires member states to provide effective enforcement procedures, including:

- Civil and administrative remedies
- Provisional measures (including border enforcement)
- Criminal procedures for willful trademark counterfeiting and copyright piracy on a commercial scale

4.3.4 The TRIPS Dispute Settlement Mechanism

A crucial innovation of TRIPS is the integration of IP disputes into the WTO dispute settlement system. WTO members can bring complaints against other members for alleged TRIPS violations, with rulings by WTO panels and the Appellate Body being binding.

Notable TRIPS disputes include:

- **India – Patent Protection for Pharmaceutical and Agricultural Chemical Products (DS50, 1997):** Challenged India's lack of product patent protection for pharmaceuticals, ultimately leading to India's patent law amendments.
- **United States – Section 110(5) of the U.S. Copyright Act (DS160, 2000):** Concerned exemptions for music performances in U.S. restaurants and bars.

4.3.5 The Doha Declaration on TRIPS and Public Health (2001)

Recognizing public health concerns in developing countries, the Doha Declaration clarified that TRIPS should not prevent members from taking measures to protect public health. It affirmed the right to use compulsory licensing and parallel importation of patented pharmaceuticals. This was followed by a 2003 waiver decision allowing export of pharmaceutical products manufactured under compulsory license.

4.4 The World Intellectual Property Organization (WIPO)

While WIPO administers many treaties (Paris, Berne, PCT, Madrid, Hague, Lisbon), it also plays a broader role:

- **Policy Coordination:** WIPO brings together stakeholders from member states, industry, and civil society to develop international IP norms.
- **Technical Assistance:** WIPO provides legislative advice and capacity building to developing countries.
- **Classification Systems:** WIPO administers the International Patent Classification (IPC) and the Nice Classification for trademarks.
- **Alternative Dispute Resolution:** WIPO offers arbitration and mediation services for IP disputes.

WIPO and WTO cooperate, particularly through a 1995 agreement, to provide technical and legal assistance on TRIPS implementation.

5. Comparative Overview of the Three Frameworks

Feature	Paris Convention	Berne Convention	TRIPS
Year	1883	1886	1994
Subject Matter	Industrial Property	Literary and Artistic Works	All IP
Administration	WIPO	WIPO	WTO
Registration Required	Yes (for patents, marks)	No (automatic)	Varies
Core Principle	National treatment, priority right	Automatic protection, national treatment	Minimum standards, MFN, enforcement
Binding Dispute Settlement	No	No	Yes (WTO mechanism)
Term	Variable	Life + 50 (minimum)	20 years for patents
Coverage	~180 countries	~180 countries	164 WTO members

6. Global Trends in Patenting

6.1 Volume and Growth

Patent applications worldwide have grown dramatically. According to WIPO data, total patent applications globally have risen from roughly 1 million in the early 2000s to over 3.5 million in recent years. This growth reflects:

- Increased R&D investment globally
- Strategic patenting by corporations
- Expansion of patent systems into emerging economies
- Growth in software, biotechnology, AI, and clean energy inventions

6.2 Geographic Concentration

Despite the global spread of patent systems, patent activity remains concentrated:

- **Top Filing Offices:** China (CNIPA), United States (USPTO), Japan (JPO), Republic of Korea (KIPO), and the European Patent Office (EPO) account for the overwhelming majority of filings.
- **China's Rise:** China overtook the United States as the largest patent filing office around 2011–2012, and the gap has widened substantially.
- **Asian Dominance:** Five of the top ten offices are now in Asia, reflecting the regional shift in innovation activity.

6.3 Sectoral Distribution

Current patenting activity is dominated by:

1. **Digital Communication and Computer Technology** (including AI/ML innovations)
2. **Electrical Machinery and Energy** (batteries, semiconductors, renewable energy)
3. **Pharmaceuticals and Biotechnology** (continuing strong activity especially around mRNA technology post-COVID)
4. **Transportation** (electric vehicles, autonomous driving)
5. **Medical Technology** (diagnostics, devices)

6.4 Trends in Computer Science Patenting

The field is particularly relevant to B.Tech CS students:

- **Software Patent Eligibility:** Remains contentious, with USPTO and EPO applying different tests. Indian patent law excludes computer programs "per se" but allows software-implemented inventions with technical effect.
- **AI-Generated Inventions:** Several jurisdictions have grappled with whether an AI can be named as inventor (notably the *Thaler v. Comptroller General* cases in the UK, EPO, and elsewhere).
- **Open Source and Patents:** Tension between open-source software licensing and patent aggression has produced initiatives like the Open Invention Network and patent non-assertion pledges.
- **Standard Essential Patents (SEPs):** Patents covering technologies incorporated into standards (e.g., 4G/5G, Wi-Fi, video codecs) raise complex FRAND (Fair, Reasonable, and Non-Discriminatory) licensing issues.

6.5 Trends in Developing Countries

- **India:** Has emerged as a significant pharmaceutical and IT patent filer; however, India has historically been more cautious about expansive patentability, particularly for pharmaceuticals and software.
- **Brazil:** Has strengthened its patent system while retaining health-related flexibilities.
- **African IP:** The establishment of the African Continental Free Trade Area (AfCFTA) protocol on IP signals movement toward a regional patent framework.
- **China:** Has moved from being primarily a patent receiving country to a major source of patent applications and international filings via PCT.

7. Harmonization of IP Laws

7.1 What is Harmonization?

Harmonization refers to efforts to reduce or eliminate differences in IP laws across jurisdictions. It exists on a spectrum:

- **Minimum Standards Harmonization:** Setting baseline obligations (the TRIPS approach).
- **Substantive Harmonization:** Aligning the actual content of laws (e.g., definitions of patentability, term, scope).
- **Procedural Harmonization:** Aligning administrative procedures and examination standards.
- **Unification:** Creating a single supranational IP right (e.g., the European Patent with Unitary Effect).

7.2 Drivers of Harmonization

1. **Trade Integration:** As goods and services cross borders, divergent IP laws create compliance costs.
2. **Technology Convergence:** Digital technologies make territorial distinctions increasingly anachronistic.
3. **Patent Quality Concerns:** Inconsistent standards (e.g., on software patentability) can lead to forum shopping.
4. **Administrative Efficiency:** Common procedures reduce duplication for applicants and offices.
5. **Anti-Counterfeiting:** Global supply chains require coordinated enforcement.

7.3 Major Harmonization Initiatives

a) The Paris Union and Strasbourg Agreement (1971, amended 1979) Established the International Patent Classification (IPC), used by patent offices worldwide.

b) Patent Law Treaty (PLT) (2000) WIPO-administered treaty harmonizing formal requirements for patent applications (filing procedures, representation, restoration of priority rights).

c) Substantive Patent Law Treaty (SPLT) Negotiations (2000–2006) WIPO negotiations aimed at harmonizing substantive patent law (definition of prior art, novelty, inventive step, grace period). These negotiations stalled due to disagreement between developed and developing countries, particularly on disclosure requirements and prior user rights.

d) European Patent Convention and Unitary Patent The European Patent Convention (1973) established the European Patent Office (EPO), granting patents that are validated nationally. The 2012–2013 package (Unitary Patent and Unified Patent Court) created a true EU-wide patent.

e) Regional Frameworks

- ARIPO (African Regional Intellectual Property Organization)
- OAPI (Organisation Africaine de la Propriété Intellectuelle)
- Eurasian Patent Organization (EAPO)
- Gulf Cooperation Council Patent Office
- The proposed ASEAN patent system

7.4 Obstacles to Harmonization

1. **National Sovereignty:** Countries jealously guard their legal traditions, particularly on issues of public health, traditional knowledge, and economic development.
2. **Diverse Legal Traditions:** Common law and civil law systems approach IP differently.

3. **Development Levels:** Developing countries often resist harmonization that they perceive as locking in standards favorable to advanced economies.
4. **Political Economy:** Powerful industries lobby for or against harmonization depending on their competitive position.
5. **Emerging Issues:** Rapid technological change (AI, blockchain, biotechnology) outpaces diplomatic processes.
6. **Enforcement Variability:** Even with harmonized laws, enforcement quality varies dramatically.

7.5 The Tension Between Harmonization and Flexibilities

A recurring theme in international IP discourse is the tension between:

- **TRIPS-plus obligations:** Bilateral or regional free trade agreements (e.g., USMCA, EU FTAs) often impose obligations beyond TRIPS, such as longer patent terms, data exclusivity, and stricter enforcement.
- **TRIPS flexibilities:** TRIPS itself contains flexibilities (compulsory licensing, parallel importation, transition periods) that developing countries seek to preserve.

The 2001 Doha Declaration emphasized that TRIPS "should be interpreted and implemented in a manner supportive of WTO members' right to protect public health and, in particular, to promote access to medicines for all."

8. Case Studies

8.1 Case Study 1: Novartis v. Union of India (2013)

The Indian Supreme Court's denial of a patent for Novartis's cancer drug Glivec (imatinib mesylate) became an international cause célèbre. The Court applied Section 3(d) of the Indian Patents Act, which prohibits patents on new forms of known substances unless they demonstrate significantly enhanced efficacy. The case:

- Demonstrated how developing countries use TRIPS flexibilities.
- Highlighted tensions between pharmaceutical innovation incentives and access to medicines.
- Drew global attention to India's careful, public-health-oriented patent examination.

8.2 Case Study 2: The Apple-Samsung Global Patent Wars

Apple and Samsung litigated smartphone patents across more than ten countries including the United States, South Korea, Germany, Japan, the UK, and Australia. The cases revealed:

- The strategic value of patents in technology markets.
- The cost of parallel litigation in multiple jurisdictions.
- The role of design patents in the smartphone wars.
- Inconsistent outcomes across jurisdictions, illustrating the case for harmonization.

8.3 Case Study 3: The COVID-19 Vaccine Patent Debate

During the COVID-19 pandemic, India and South Africa proposed a TRIPS waiver to suspend certain IP protections on COVID-19 vaccines, therapeutics, and diagnostics. The proposal, ultimately narrowed to vaccines and adopted in 2022 after lengthy negotiations, demonstrated:

- The use of international IP frameworks in public health crises.
 - Deep divisions between developed and developing economies.
 - The role of IP in global access to essential medicines.
 - The intersection of IP, trade, and global health governance.
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9. Implications for Computer Science Research

For B.Tech CS students engaged in research, understanding the international IP landscape is critical for several reasons:

1. **Filing Decisions:** Students developing novel algorithms, AI models, or software should consider international filing strategy (priority rights, PCT).
 2. **Publication vs. Patenting:** Most jurisdictions require patent filing before any public disclosure. Conference papers, GitHub releases, and thesis submissions can destroy novelty.
 3. **Open Source Compliance:** Understanding how patents interact with GPL, MIT, Apache, and other open-source licenses is essential.
 4. **Standard Setting:** Participation in standards bodies (IEEE, IETF, W3C) requires awareness of SEP obligations.
 5. **Technology Transfer:** Researchers collaborating with international partners must understand ownership and licensing across jurisdictions.
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10. Summary

This lecture surveyed the international landscape of intellectual property, focusing on three foundational instruments:

- The **Paris Convention** (1883) established the principle of national treatment and the right of priority for industrial property rights.
- The **Berne Convention** (1886) created automatic, formalities-free protection for literary and artistic works, including moral rights.
- The **TRIPS Agreement** (1994) transformed the global IP landscape by incorporating minimum standards and binding dispute settlement within the WTO framework.

We examined global patenting trends, noting the geographic shift toward Asia, the rise of computer science and digital technology patenting, and the emergence of novel legal questions around AI, software, and standards-essential patents.

Finally, we analyzed ongoing efforts to harmonize IP laws, recognizing both the practical benefits of convergence and the principled objections rooted in national sovereignty and development priorities. The tension between TRIPS-plus standards and TRIPS flexibilities remains a defining feature of contemporary international IP politics.

For computer science researchers, the international IP system is not a distant abstraction but a practical framework that shapes how innovations are protected, shared, and commercialized across borders.

11. Exam Preparation Tips

11.1 Key Concepts to Memorize

- Founding dates and original signatories of Paris and Berne Conventions
- The 12-month and 6-month priority periods under Paris
- The three-step test under Berne Article 9(2)
- The 20-year minimum patent term under TRIPS Article 33
- The seven IP domains covered by TRIPS
- The Doha Declaration principles
- Differentiating between national treatment and most-favoured-nation treatment

11.2 Common Exam Question Patterns

1. **Comparative questions:** "Compare and contrast the Paris Convention and the Berne Convention with respect to scope, principles, and protection mechanism."
2. **Application questions:** "An inventor in India files a patent application on 1 March. By what date must she file corresponding applications in other Paris Convention states to preserve priority?"

3. **Critical analysis:** "Discuss the tension between TRIPS flexibilities and TRIPS-plus obligations in bilateral free trade agreements."
4. **Case-based questions:** "Analyze the Novartis v. Union of India decision in light of TRIPS obligations."
5. **Short notes:** "TRIPS and Public Health" / "Right of Priority" / "Three-Step Test" / "Harmonization of IP Laws."

11.3 Study Strategy

- Create a comparative table covering each major framework (Paris, Berne, TRIPS) along dimensions such as year, subject matter, administration, registration, key principles, and dispute settlement.
 - Practice linking treaty articles to specific real-world scenarios.
 - Read at least one WIPO or WTO official resource on each treaty.
 - Discuss with peers the implications of recent cases and policy debates.
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12. Practice Questions

12.1 Short Answer Questions

1. What is the right of priority under the Paris Convention, and why is it commercially important?
2. Differentiate between automatic protection under the Berne Convention and registration-based protection under the Paris Convention.
3. List the seven categories of IP covered by TRIPS.
4. What is the three-step test, and in which treaty article is it found?
5. State the minimum term of patent protection under TRIPS.

12.2 Long Answer Questions

1. Critically examine the contribution of the TRIPS Agreement to the international IP system. How does it differ from the Paris and Berne Conventions in scope and enforcement?
2. "The Paris Convention facilitates the acquisition of patent rights but does not provide an international patent." Discuss this statement in the context of subsequent international instruments.
3. Discuss the major global trends in patent filing activity over the past two decades. What do these trends suggest about the geography of innovation?

4. Examine the arguments for and against the harmonization of intellectual property laws. Illustrate your answer with reference to specific initiatives such as the SPLT and the European Unitary Patent.
5. Analyze the significance of the Doha Declaration on TRIPS and Public Health. How have subsequent developments, including the COVID-19 vaccine waiver, reflected the principles of the Declaration?

12.3 Application-Based Questions

1. **Scenario:** Dr. Aisha, a researcher based in Egypt, files a patent application on 15 April 2024. She wishes to file corresponding applications in Germany, Japan, and the United States. By what dates must she file in each jurisdiction to preserve her Paris Convention priority? What happens if she misses the deadline for the United States but files in Germany on time?
 2. **Scenario:** A software development company in India has developed an innovative AI-based medical diagnostic tool. The CTO is debating whether to seek patent protection in foreign markets or rely on trade secret protection. Advise the company, considering relevant international IP frameworks.
 3. **Scenario:** The government of Country X is considering whether to issue a compulsory license for a patented HIV antiretroviral drug to address a public health crisis. Analyze the legality and implications of this action under TRIPS and the Doha Declaration.
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13. Suggested Further Reading

- WIPO Official Website: <https://www.wipo.int> — Treaty texts, statistics, and policy resources.
 - WTO TRIPS Text and Documents: https://www.wto.org/english/tratop_e/trips_e/trips_e.htm
 - World Intellectual Property Indicators (Annual WIPO Publication) — Global patenting statistics.
 - Peter Drahos and John Braithwaite, *Information Feudalism: Who Owns the Knowledge Economy?* (Earthscan, 2002).
 - Carlos Correa, *Trade Related Aspects of Intellectual Property Rights: A Commentary on the TRIPS Agreement* (Oxford University Press, 2007).
 - UNCTAD, *Resource Book on TRIPS and Development* (Cambridge University Press, 2005).
 - Jerome Reichman and others, *International Intellectual Property Law and Policy* (various editions).
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End of Lecture 5 Notes